Lohit Gali

Senior Data Engineer
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Summary

- Senior Data Engineer with 5+ years of experience designing, building, and operating enterprise data warehouses and analytical data platforms in regulated, data-intensive environments.
- Proven track record of developing reliable batch and near-real-time data pipelines, integrating third-party data providers, and implementing data quality, reconciliation, and surveillance frameworks that support business-critical reporting and analytics.
- Strong collaborator with business, analytics, and engineering teams, translating complex domain requirements into scalable, governed data solutions.

Skills

Data Warehousing & Analytics: Enterprise Data Warehouse (EDW), dimensional modeling, fact & dimension tables, historical trend analysis, time-series data, reconciliation datasets
Data Engineering: Azure Databricks, PySpark, Spark SQL, Delta Lake, Delta Live Tables, ETL / ELT pipelines, batch & incremental loads, schema evolution, data validation, idempotent processing
Data Quality & Surveillance: Data completeness checks, anomaly detection, freshness monitoring, control totals, audit trails
Programming: Python, SQL (T-SQL, Spark SQL)
APIs & Third-Party Data: REST APIs, pagination, rate-limit handling, incremental ingestion patterns
DevOps & Reliability: CI/CD for data pipelines, version control (Git), logging, error handling

Professional Experience

CIFC Asset Management *July 2025 – Present*
Senior Data Engineer

- Designed and enhanced an enterprise analytical data warehouse to consolidate operational, security, and third-party datasets, enabling long-term analytics beyond source-system retention limits.
- Built ETL/ELT pipelines using PySpark and Delta Lake to ingest, normalize, and historize structured and semi-structured data for downstream reporting and analytics.
- Implemented incremental ingestion frameworks for external SaaS and vendor APIs lacking native CDC, using watermarking, idempotent writes, and replay-safe logic to ensure data accuracy.
- Developed data surveillance and quality controls including row-count reconciliation, freshness checks, schema validation, and anomaly detection to proactively identify data issues impacting reports.
- Optimized analytical tables through partitioning, clustering, and file compaction, reducing query latency and improving performance for business-facing dashboards.
- Productionized pipelines with workflow orchestration, adding monitoring, alerting, and failure-handling to meet daily SLA requirements.
- Partnered closely with reporting, analytics, and business stakeholders to translate analytical requirements into well-modeled, consumable datasets.

Samay Systems LLC *Feb 2025 – Jun 2025*
Senior Data Engineer

- Architected a centralized data warehouse layer to support a compliance and analytics SaaS platform, integrating multiple internal and third-party data sources.

- Designed ELT pipelines to transform raw ingestion layers into analytics-ready datasets with consistent schemas and historical traceability.
- Implemented data validation and control checks at ingestion and transformation stages to prevent downstream data integrity issues.
- Enabled reliable deployments by integrating data pipelines into CI/CD workflows, ensuring versioned, repeatable releases across environments.
- Worked directly with product and customer teams to align data models with reporting, audit, and analytical use cases.

Arvam Tech *Jul 2023 – Jan 2025*
Data Engineer

- Built batch ETL pipelines to consolidate enterprise asset and inventory data into a centralized data lake for long-term analytics.
- Developed Python and Spark SQL transformations to standardize schemas, enforce referential integrity, and support cross-system reporting.
- Implemented incremental load strategies for API-based sources lacking CDC, minimizing data movement and processing cost.
- Implemented pipeline monitoring and alerts to track data freshness, failures, and SLA compliance.
- Partnered with stakeholders to evolve data models as business requirements changed, maintaining backward compatibility.

RHYM Tech *Aug 2022 – May 2023*
Data Engineer

- Developed high-volume data ingestion pipelines to process telemetry and event data into analytical storage for historical analysis.
- Built optimized analytical tables to support trend analysis, investigation workflows, and reporting.
- Tuned Spark jobs by optimizing joins, partitions, and file sizes to improve pipeline reliability and performance.

ZeMoSo Technologies *Jul 2021 – Aug 2022*
Associate Developer

- Implemented data ingestion and transformation logic using Python and SQL to support scientific and operational workflows.
- Assisted in designing structured data models aligned with reporting and analytics requirements.
- Supported production deployments with logging, monitoring, and basic data quality checks.

Projects

Enterprise Data Warehouse Modernization
Problem: Fragmented systems and short data retention windows limited analytics and trend analysis.
Solution: Built governed Azure Databricks pipelines using PySpark, Delta Lake, and Unity Catalog to centralize, secure, and scale analytical data with SLA-driven reliability.
Impact: Enabled consistent reporting, trend analysis, and improved confidence in business metrics.

Agent-based Compliance Dashboard – RHYM Tech
Problem: External vendor data arrived inconsistently and caused reporting discrepancies.
Solution: Built incremental ingestion pipelines with reconciliation checks, anomaly detection, and freshness monitoring.
Impact: Reduced data issues reaching dashboards and improved trust in analytical outputs.

Education

Master's in Computer Science, Oklahoma State University, Stillwater, OK *May 2024*
Bachelor's in Computer Science, JNTU Hyderabad, India *Jun 2021*